Productivity and programme assumptions report

DEMONSTRATION EXPORT - NOT SUBMISSION READY - NOT APPROVED FOR ANY REAL TENDER | approvals by: test principals in environment 'test'

SYNTHETIC TEST DATA - FICTIONAL CUSTOMER - NOT A REAL TENDER - NOT A MARKET QUOTE - NOT FOR SUBMISSION

Project ID	PRJ-9101
Project name	Dammam Industrial Utilities - Synthetic Tender Closure
Customer	Gulf Process Utilities Co. (FICTIONAL SYNTHETIC CUSTOMER)
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Title	Productivity and programme assumptions report
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Reviewer	not yet reviewed
Estimate version	v2
Estimate fingerprint	48d0f9242dc910b7
Template	T-PRODUCTIVITY@1.0.0
Confidentiality	CONFIDENTIAL (synthetic test data) - no real customer or supplier data

Productivity norms used

Norm	Ver	Trade	Activity	Basis	Value	Unit	Status	Evidence	Origin	Source	Flag
N-BACK	1	Civil	Backfill and compaction	crew_output_per_crew_day	80	m3	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-CABLE	1	Electrical	Cable pulling	crew_output_per_crew_day	120	m	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic

Norm	Ver	Trade	Activity	Basis	Value	Unit	Status	Evidence	Origin	Source	Flag
N-COAT	1	Piping	Coating	crew_output_per_crew_day	60	m2	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-EXC	1	Civil	Trench excavation	crew_output_per_crew_day	60	m3	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-EXCH	1	Civil	Excavation in hard material	crew_output_per_crew_day	12	m3	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-FIBRE	1	ELV	Fibre backbone	crew_output_per_crew_day	100	m	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-FORM	1	Structural	Formwork	crew_output_per_crew_day	20	m2	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-NDT	1	Piping	Radiographic testing	crew_output_per_crew_day	6	joint	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic

Norm	Ver	Trade	Activity	Basis	Value	Unit	Status	Evidence	Origin	Source	Flag
N-PIPE	1	Piping	Pipe fabrication and installation	labour_hours_per_unit	0.5	inch_dia	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-POUR	1	Structural	Concrete pour	crew_output_per_crew_day	24	m3	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-PUMP	1	Mechanical	Pump skid setting	crew_output_per_crew_day	0.5	nr	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-REBAR	1	Structural	Reinforcement fix	labour_hours_per_unit	0.04	kg	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-ROAD	1	Civil	Road reinstatement	crew_output_per_crew_day	45	m2	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-SHORE	1	Civil	Trench shoring	crew_output_per_crew_day	40	m	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic

Norm	Ver	Trade	Activity	Basis	Value	Unit	Status	Evidence	Origin	Source	Flag
N-STEEL	1	Structural	Steel erection	crew_output_per_crew_day	2500	kg	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-SUPP	1	Piping	Support fabrication	crew_output_per_crew_day	500	kg	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-TRAY	1	Electrical	Cable tray	crew_output_per_crew_day	45	m	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic
N-WELD	1	Piping	Butt welding	labour_hours_per_unit	3	joint	Approved	synthetic	synthetic	SYNTHETIC TEST DATA - NOT AN INDUSTRY BENCHMARK - NOT A MARKET QUOTE	synthetic

Gaps

Activity	Kind	Detail
A-EXC	NORM_EVIDENCE_SYNTHETIC	norm N-EXC v1: evidence 'synthetic'
A-EXCH	NORM_EVIDENCE_SYNTHETIC	norm N-EXCH v1: evidence 'synthetic'
A-SHORE	NORM_EVIDENCE_SYNTHETIC	norm N-SHORE v1: evidence 'synthetic'
A-FORM	NORM_EVIDENCE_SYNTHETIC	norm N-FORM v1: evidence 'synthetic'
A-REBAR	NORM_EVIDENCE_SYNTHETIC	norm N-REBAR v1: evidence 'synthetic'
A-POUR	NORM_EVIDENCE_SYNTHETIC	norm N-POUR v1: evidence 'synthetic'
A-STEEL	NORM_EVIDENCE_SYNTHETIC	norm N-STEEL v1: evidence 'synthetic'
A-SUPP	NORM_EVIDENCE_SYNTHETIC	norm N-SUPP v1: evidence 'synthetic'

Activity	Kind	Detail
A-PUMP	NORM_EVIDENCE_SYNTHETIC	norm N-PUMP v1: evidence 'synthetic'
A-PIPE100	NORM_EVIDENCE_SYNTHETIC	norm N-PIPE v1: evidence 'synthetic'
A-PIPE50	NORM_EVIDENCE_SYNTHETIC	norm N-PIPE v1: evidence 'synthetic'
A-WELD	NORM_EVIDENCE_SYNTHETIC	norm N-WELD v1: evidence 'synthetic'
A-NDT	NORM_EVIDENCE_SYNTHETIC	norm N-NDT v1: evidence 'synthetic'
A-BACK	NORM_EVIDENCE_SYNTHETIC	norm N-BACK v1: evidence 'synthetic'
A-ROAD	NORM_EVIDENCE_SYNTHETIC	norm N-ROAD v1: evidence 'synthetic'
A-COAT	NORM_EVIDENCE_SYNTHETIC	norm N-COAT v1: evidence 'synthetic'
A-TRAY	NORM_EVIDENCE_SYNTHETIC	norm N-TRAY v1: evidence 'synthetic'
A-CABLE	NORM_EVIDENCE_SYNTHETIC	norm N-CABLE v1: evidence 'synthetic'
A-ELV	NORM_EVIDENCE_SYNTHETIC	norm N-FIBRE v1: evidence 'synthetic'

Method

Required labour-hours = quantity x approved labour-hours per unit; crew-days = hours / (productive hours per worker x crew size); duration = ceil(crew-days / effective crews). Crew-output norms use units per crew-day and do not re-apply the labour-hours formula.

Assumptions, exclusions and limitations

- Quantities are human-measured BOQ quantities; the application does not measure drawings.
- Figures are taken from frozen estimate version v2 (fingerprint 48d0f9242dc910b7).
- SYNTHETIC / DEMONSTRATION data: rates, quotations, productivity norms, customer and responses are fictional test inputs, not market evidence.
- No norm is an industry benchmark; every value is a sourced company/project entry.

Source references

- Frozen estimate version v2 (fingerprint 48d0f9242dc910b7), totals and pricing schedule
- Registers in the application database (see document D33 for received documents, D08 for assumptions, D25 for queries)